Economic features and selection risk in daily USD/JPY forecasting

Sai Reddivari

This companion records the formulas, operational conventions and saved results behind the main study. It is optional reading, not a second experiment. No models were fitted or scored for this public adaptation. Definitions distinguish theoretical mechanisms from the library settings actually used. The underlying project originated in applied CQF research; course materials and the exam paper are not redistributed.

1. Target, clocks and transformations

The target uses simple returns, while several momentum features use log returns. With $r_t = P_t/P_{t-1} - 1$,

$$\bar{r}_{20,t} = \frac{1}{20}\sum_{j=0}^{19} r_{t-j}, \quad \widehat{\sigma}^2_{20,t} = \frac{1}{19}\sum_{j=0}^{19}(r_{t-j} - \bar{r}_{20,t})^2, \quad Y_t = \mathbf{1}\{r_{t+1} > \max(0.001, 0.5\widehat{\sigma}_{20,t})\}.$$

The window includes the return into t, not the future outcome. A Friday decision runs to the next retained session, not necessarily a 24-hour horizon. No current observation is removed simply because the future return is near zero.

Under a temporary illustration $r_{t+1} \mid \mathcal{F}_t \sim N(\mu_t, \sigma_t^2)$,

$$\Pr(Y_t = 1 \mid \mathcal{F}_t) = 1 - \Phi\left(\frac{\max(0.001, 0.5\widehat{\sigma}_{20,t}) - \mu_t}{\sigma_t}\right).$$

If the floor is inactive, estimated and conditional volatility coincide and drift is small, this is approximately $1 - \Phi(0.5) = 0.309$. It motivates scaling, not an assumption that class rates are constant. The floor binds on one of 1,277 labelled development rows and 21 of 65 reserved R3 decisions, with no binding rows in the other reserved quarters.

Native transformations precede the as-of join

Write $\ell_k(A)_d = \log A_d - \log A_{d-k}$ on asset A's native observations. If $\log A_d = \log A_{d-1} + \mu + \epsilon_d$ with stationary increments, $\ell_1(A)_d = \mu + \epsilon_d$ is stationary. The premise is essential: this argument does not establish stationarity for the observed financial series, volatility, levels, cross-asset spreads or rolling statistics. Fractional differentiation was not used.

For a source with availability time a_j, the decision uses $j^*(t) = \max\{j : a_j \le d_t\}$ in chronological availability order. Carrying an old observation forward is not a new measurement. Native-session returns and weekly-report differences are computed before that join; aligned US–Japan gaps use the latest eligible legs, retaining both leg dates.

Trailing standardization is $z_n(x)_t = (x_t - \bar{x}_{n,t})/s_{n,t}$, with sample SD, current available observation included, full warm-up required and denominator at least 10^{-12}. Positioning uses 52 distinct reports; daily volatility and interaction components use 260 decision rows. Fold-fitted scaling for elastic net is a different operation, learned on each inner training set alone.

Source-specific availability conventions

Input / source	Assigned availability	Qualification
FX / FMP	Daily bar assigned to 17:00 New York	Provisional boundary; close-derived features and labels inherit it. Basket uses the same source/clock.
US ETFs / FMP	16:05 New York	Five-minute buffer after regular-session close; historical delivery timestamps are not verified.
VIX / FRED-Cboe	16:20 New York	Adopted 16:15 observation plus buffer.
US yields / FRED	16:30 New York	Adopted afternoon availability, not a verified provider latency log.
Japanese sectors / J-Quants	15:35 JST	Tokyo close changed from 15:00 to 15:30 on 5 November 2024; conservative visibility convention and native-calendar construction.
Japanese yields / MOF	15:05 JST	Market-visibility assumption for 15:00 observations, rather than the later MOF posting time.
BOJ FX reference	17:05 JST	Adopted visibility of 17:00 midpoint; reference-to-close move spans Europe as well as New York.
CFTC yen reports	Actual release date, 15:35 New York	Observation date and release date differ. Original age cap: 35 days from observation.
WTI / FRED-EIA	Previous native observation, released with next observation's adopted 15:00 New York time	Conservative one-observation lag; historical posting time unverified.
Event calendar	Schedule known by decision cutoff	No announcement result. Some rescheduling timing remains unresolved and is flagged.

New York time uses daylight-saving conversion; Tokyo remains JST. Latest-vintage downloads and visibility conventions are not a point-in-time database. Four reserved event flags exist, but three lie in the original positioning exclusion: removing the one remaining flagged row from the 229-row primary sample changes mean AUC by at most about 0.0052. The final training fit excludes nine flagged development dates.

2. Selection objectives and executed rules

Structural filtering and the light marginal screen

Stage A checks finite, nonmissing values, removes constants and byte-identical columns, and clusters by distance $d_{ij} = 1 - |\rho^S_{ij}|$ with complete linkage at 0.10. Representatives follow declared conditional preferences; otherwise the feature with highest mean absolute within-cluster correlation is retained. The rule uses the outer training window and no label. Components can remain available for constructions even if not retained as standalone inputs.

Stage B estimates mixed discrete/continuous mutual information:

$$I(X;Y) = \sum_y \int p(x,y) \log \frac{p(x,y)}{p(x)p(y)}\, dx.$$

It uses `mutual_info_classif`, three neighbours and seed zero; event/weekend flags are declared discrete [8, 6]. For p eligible columns, the lowest $\lfloor p/4 \rfloor$ ranks in each of three inner training folds are marked. A column is removable only if it is in this bottom group in all three folds. Removal proceeds in increasing mean MI, protecting the final eligible member of a family and declared interactions/components. This is a light ranking rule, not a hypothesis test against a shifted-label cutoff.

The selector union preserves complementary routes

The linear route standardizes on inner training data and fits logistic elastic net. With $p_i = \sigma(b + x_i^\top \beta)$ and $\rho = 0.5$, its equivalent mean-loss objective for unit weights is

$$J(b, \beta) = \frac{1}{n} \sum_i [-y_i \log p_i - (1 - y_i) \log(1 - p_i)] + \frac{1}{nC} \left[\frac{1 - \rho}{2} \|\beta\|_2^2 + \rho \|\beta\|_1 \right].$$

The intercept is unpenalized. $C \in \{0.01, 0.03, 0.1, 0.3, 1\}$ is chosen by mean inner AUC, with ties favouring smaller C; a candidate has a nonzero coefficient in at least two of three fits. Solver: SAGA, tolerance 10^{-4}, maximum 20,000 iterations [7]. The yen-relative contrast is held out of this linear route because it equals own five-day return minus the dollar basket; the independent basis retains those two components. Trees can still see the contrast.

The nonlinear route uses LightGBM with seven leaves, depth cap three, minimum nominal leaf size 40, learning rate 0.03, L2 of one and 80% row/column sampling. Run A uses AUC early stopping (cap 600, patience 50). B, C and final use 150 fixed trees. Grouped permutation importance is

$$D_{g,k} = \frac{1}{20} \sum_{b=1}^{20} \left[AUC(y_k, \hat{p}_k) - AUC(y_k, \hat{p}_k^{\pi_{g,b}}) \right].$$

All members of group g share the same row permutation in a repeat. A group enters with positive mean drop in at least two inner folds. Permutation breaks feature relationships and is a selection heuristic, not a causal intervention. The union of linear/nonlinear candidates is supplemented by the declared interactions and available components.

The wrapper and why a chronological score can still be optimized to noise

For subset S, the wrapper scorer is $\widehat{A}(S) = \frac{1}{3} \sum_{k=1}^{3} AUC_k(S)$. At each step it tests single-column and applicable group removals and follows the best mean score. Cached subsets are reused. A new candidate consumes three fits; the path stops at the budget or a single feature. Across all evaluated subsets $\mathcal{E}$, the selected subset is

$$S^* = \arg\min_{S \in \mathcal{E}} |S| \quad \text{subject to} \quad \widehat{A}(S) \geq \max_{U \in \mathcal{E}} \widehat{A}(U) - 0.005,$$

with higher score breaking size ties. Thus 0.005 is a subset-selection tolerance, not a significance threshold or an early stopping threshold for the path. Budgets were 400 fits in A and 2,000 in B. The three declared interaction comparisons were outside that wrapper cap. C and final omit both wrapper and separate interaction comparisons.

The same inner quarters support repeated choices. As the number of candidates grows, $\max_S \widehat{A}(S)$ can increase because of estimation error rather than true improvement. Purging prevents label overlap; it does not remove this adaptive search optimism. Development comparisons and the shifted-label diagnostics examine that distinction.

Negative-control scope

Twenty circular shifts were built from $k_j = \text{round}(300 + 677j/19)$ for $j = 0, \ldots, 19$, replacing the closest available entries with anchors 319, 426 and 639. The shift was applied to the complete 1,277-row label vector; minimum circular separation was at least 300. The reduced B chain uses its selection model for final prediction, not the full three-booster grids. Real and shifted statistics are compared only within the same procedure. The public CSV retains all three diagnostic summaries. Circular shifts are not exchangeable draws from a fully justified null, so exceedance counts are not reported as calibrated p-values.

3. Boosting mathematics and actual architecture choices

For margin F, binary logistic loss is $\ell(y, F) = \log(1 + e^F) - yF$, with $g = p - y$ and $h = p(1 - p)$, where $p = \sigma(F)$. Adding tree $f(x) = w_{q(x)}$ gives the second-order approximation

$$\widetilde{\mathcal{L}} = \sum_i [g_i f(x_i) + \tfrac{1}{2} h_i f(x_i)^2] + \gamma T + \alpha \sum_{j=1}^{T} |w_j| + \tfrac{\lambda}{2} \sum_{j=1}^{T} w_j^2.$$

For leaf j, write $G_j = \sum_{i \in I_j} g_i$, $H_j = \sum_{i \in I_j} h_i$ and soft threshold $\mathcal{S}_\alpha(G) = \operatorname{sgn}(G) \max(|G| - \alpha, 0)$. Assuming $H_j + \lambda > 0$,

$$w_j^* = -\frac{\mathcal{S}_\alpha(G_j)}{H_j + \lambda}, \qquad \widetilde{\mathcal{L}}^* = -\frac{1}{2} \sum_j \frac{\mathcal{S}_\alpha(G_j)^2}{H_j + \lambda} + \gamma T.$$

The canonical gain for splitting parent P is

$$\frac{1}{2} \left[\frac{\mathcal{S}_\alpha(G_L)^2}{H_L + \lambda} + \frac{\mathcal{S}_\alpha(G_R)^2}{H_R + \lambda} - \frac{\mathcal{S}_\alpha(G_P)^2}{H_P + \lambda} \right] - \gamma.$$

L1 can zero a leaf update; L2 shrinks it and changes split valuation. Shrinkage multiplies added-tree output by the learning rate after fitting. These equations describe a common second-order formulation, not an assertion of identical normalization or splitting criteria in every library.

The recorded XGBoost/LightGBM implementations compare an unhalved score, so raw split penalties must be interpreted on their code scale. LightGBM picks the current leaf with the largest available split gain; configured XGBoost uses depth-wise expansion with histogram splits. XGBoost also supports other growth policies, which were not used here [4, 10].

CatBoost uses symmetric trees: one split condition at each depth is applied to all current nodes. The recorded CPU split criterion is Cosine rather than the Hessian gain above. With negative-gradient sum D_j, scoring weight W_j and regularized average $a_j = D_j/(W_j + \lambda_{\text{scaled}})$, its deterministic score is proportional to

$$\frac{\sum_j a_j D_j}{\sqrt{\sum_j a_j^2 W_j}}.$$

Split-score noise is controlled by `random_strength`; leaf values subsequently use regularized Newton estimation. Effective defaults matter: expanded CatBoost used ten leaf-estimation iterations, own-price CatBoost one [2, 1].

Ordered statistics explain an architecture, not these results

For category c_i, an ordered target statistic can use a permutation π:

$$\widehat{x}_i = \frac{\sum_{j:\pi(j)<\pi(i)} \mathbf{1}\{c_j = c_i\} y_j + a\, p_0}{\sum_{j:\pi(j)<\pi(i)} \mathbf{1}\{c_j = c_i\} + a}.$$

Excluding y_i prevents its direct encoding into its own feature. Ordered boosting applies related ordering to residual estimation; these are distinct mechanisms [5]. Neither supplies a calendar-time availability guarantee. This study used numeric features and Plain boosting, so no categorical target statistic or Ordered boosting explains the results. The constructor passed `has_time=True`; its absence in fitted read-back was documented, and no beneficial effect was inferred.

4. Tuning, refit and threshold specification

Model	Grid (Cartesian product: sixteen configurations from B onward)
LightGBM	leaves $\{7, 15\}$; nominal minimum child rows $\{20, 40\}$; learning rate $\{0.03, 0.1\}$; minimum split gain $\{0, 0.1\}$. L2 fixed at one; 80% row/column sampling.
XGBoost	depth $\{3, 4\}$; minimum child Hessian sum $\{1, 5\}$; learning rate $\{0.03, 0.1\}$; gamma $\{0, 1\}$. L2 one; histogram, depth-wise growth; 80% row/column sampling.
CatBoost	depth $\{4, 6\}$; L2 $\{3, 10\}$; learning rate $\{0.03, 0.1\}$; random strength $\{1, 3\}$. Symmetric trees; Bernoulli 80% row sampling; no column subsampling.

Run A uses the first three dimensions with the extra knob at default. Grids are coarse sample-size-driven choices, not assertions that an optimum is contained. XGBoost's Hessian floor differs from a row-count floor: $H_j = \sum_i p_i(1 - p_i) \leq n_j/4$, so $H_j \geq w$ implies at least $4w$ rows and usually more. LightGBM approximates nominal counts through Hessian calculations; its minimum row parameter is not an exact universal count bound. Symmetric CatBoost does not apply the same per-leaf minimum-row mechanism [3, 9, 1].

For every configuration: fit each inner training block with AUC early stopping on its own validation block, cap 1,000 trees, patience 50. Select highest mean inner AUC, ties to the first grid entry. Refit on the complete purged training window using the chosen configuration's tree count from the latest inner fold. No outer or reserved observations select it. Thresholds are midpoints between distinct pooled inner probabilities, maximizing balanced accuracy; first maximum wins, with 0.5 fallback for fewer than two distinct probabilities. Training loss, tuning AUC and classification threshold objective are different quantities.

Procedure	Configuration	Trees	Inner AUC	τ
LightGBM: shortlist	leaves 15; rows 20; rate 0.1; gain 0.1	5	0.638	0.333
XGBoost: shortlist	depth 3; weight 1; rate 0.1; gamma 1.0	128	0.612	0.335
CatBoost: shortlist	depth 4; L2 10; rate 0.1; strength 3	18	0.653	0.342
LightGBM: own price	leaves 15; rows 40; rate 0.03; gain 0.0	7	0.636	0.319
XGBoost: own price	depth 3; weight 5; rate 0.1; gamma 1.0	15	0.621	0.318
CatBoost: own price	depth 4; L2 3; rate 0.03; strength 1	5	0.597	0.472

Frozen metadata only; no model binaries or training observations are included. Counts come from the latest inner fold, not a reserved-year optimization.

Recorded environment: Python 3.14.8, NumPy 2.5.3, pandas 3.0.6, scikit-learn 1.9.1, LightGBM 4.7.0, XGBoost 3.4.1, CatBoost 1.2.10 and SHAP 0.54.1. The Python modules preserve these recorded version checks; a newer environment is not silently treated as identical. The metadata and aggregate values were read from saved artifacts; the public adaptation does not reload models.

5. Metrics and uncertainty

For positive/negative counts n_+, n_- and scores s_i, empirical AUC is

$$\frac{1}{n_+ n_-} \sum_{i:y_i=1} \sum_{j:y_j=0} \left[\mathbf{1}\{s_i > s_j\} + \tfrac{1}{2}\mathbf{1}\{s_i = s_j\} \right].$$

It measures ranking and is invariant to strictly increasing score transformations. Calibration and fixed-threshold decisions need not be invariant. With confusion counts TP, FP, TN and FN,

$$\text{precision} = \frac{TP}{TP+FP}, \quad \text{recall} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2TP}{2TP+FP+FN},$$

$$\text{accuracy} = \frac{TP+TN}{N}, \qquad \text{BA} = \frac{1}{2}\left[\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right].$$

Class-0 metrics swap the positive/negative roles; macro averages weight classes equally and weighted averages use support. Undefined divisions follow the saved zero-division convention. The archived identifier `majority` denotes a constant training-prevalence probability, not an always-majority-class decision rule. Its frozen threshold predicts all primary reserved rows positive.

The primary aggregation is the equal-quarter mean; SD uses denominator $K-1$. Pooled AUC also compares ranks across quarters and can differ from the mean, especially when model score scales change across development refits. Calibration bins are the saved five quantile bins, not a recalibration exercise. Reference independent-observation AUC standard errors are not used to claim paired time-series significance. No profitability, Sharpe ratio or drawdown is inferred.

6. Complete candidate dictionary and final membership

The following 39 definitions reproduce the saved manifest. The final 34 exclude `ret_1d`, `ret_20d`, `vol_5_over_20`, `dist_ma20` and `us_slope_chg_5d`, all through the light MI screen. Native-calendar index d, report index j and aligned decision index t are distinguished. ℓ_k is a native log return, s_n a sample SD and z_n a trailing standardized value. Four sector features depend on licensed J-Quants observations; none of those observations are reproduced here. The ten own-price baseline features remain the entire currency-state family, including features removed from the expanded model.

Own-market state

Feature / final status	Definition and hypothesis
`ret_1d` Removed by MI	$\ln(P_t / P_{t-1})$ Last-session momentum or reversal. *Calendar:* Retained FX sessions
`ret_5d` Retained	$\ln(P_t / P_{t-5})$ Short trend in the currency quote. *Calendar:* Retained FX sessions
`ret_20d` Removed by MI	$\ln(P_t / P_{t-20})$ Medium trend in the currency quote. *Calendar:* Retained FX sessions
`abs_ret_1d` Retained	$\lvert\ln(P_t / P_{t-1})\rvert$ Size of the latest move; may describe changing volatility. *Calendar:* Retained FX sessions
`vol_20` Retained	$s_{20}(r)_t = \sigma_{20,t}$ Trailing volatility sets the scaled part of the label hurdle. This is not the full hurdle when the floor binds. *Calendar:* Retained FX sessions
`vol_5_over_20` Removed by MI	$s_5(r)_t / s_{20}(r)_t$ Recent volatility relative to its longer baseline. *Calendar:* Retained FX sessions
`dist_ma20` Removed by MI	P_t / mean(P_{t-19}, …, P_t) - 1 Distance from a trailing price average; a possible stretch or trend state. *Calendar:* Retained FX sessions
`dist_ma60` Retained	P_t / mean(P_{t-59}, …, P_t) - 1 Distance from a trailing price average; a possible stretch or trend state. *Calendar:* Retained FX sessions
`tokyo_reference_to_ny_close` Retained	P_t / $BOJfix_{latest}$ - 1 Move since the Tokyo reference, including European trading; compares an interbank midpoint with a vendor composite close. *Calendar:* FX decisions + Tokyo reference
`weekend_in_window` Retained	$\mathbf{1}\{(t, t+1]$ spans a Saturday or Sunday$\}$ Scheduled exposure to weekend news, without future market information. *Calendar:* Retained FX decision calendar

Relative rates

Feature / final status	Definition and hypothesis
`gap2y` Retained	$US2y_{latest}$ - $JGB2y_{latest}$ Relative yield level as policy and carry context. *Calendar:* US and Tokyo legs aligned separately

Feature / final status	Definition and hypothesis
`gap2y_chg_1d` Retained	$gap2y_t$ - $gap2y_{t-1}$ Change in the aligned rate differential as a policy-repricing hypothesis. *Calendar:* Decision rows
`gap2y_chg_5d` Retained	$gap2y_t$ - $gap2y_{t-5}$ Change in the aligned rate differential as a policy-repricing hypothesis. *Calendar:* Decision rows
`us_slope_chg_5d` Removed by MI	$(US10y - US2y)_d$ - $(US10y - US2y)_{d-5}$ Change in the US curve as a policy and growth-expectations hypothesis. *Calendar:* US Treasury sessions
`jp_slope_chg_5d` Retained	$(JGB10y - JGB2y)_d$ - $(JGB10y - JGB2y)_{d-5}$ Change in the Japanese curve as a BOJ-expectations hypothesis. *Calendar:* Tokyo MOF sessions

Japanese sectors

Feature / final status	Definition and hypothesis
`banks_rel_5d` Retained	ℓ_5(Banks) - ℓ_5(TOPIX) Banks relative to the Japanese market as a domestic-rate hypothesis. *Calendar:* Tokyo source sessions
`banks_rel_20d` Retained	ℓ_{20}(Banks) - ℓ_{20}(TOPIX) Slower bank-sector relative performance. *Calendar:* Tokyo source sessions
`elec_rel_5d` Retained	ℓ_5(Elec & Precision) - ℓ_5(TOPIX) Broad electronics relative performance; not a semiconductor or exporter basket. *Calendar:* Tokyo source sessions
`topix_5d` Retained	ℓ_5(TOPIX) Broad Japanese equity risk context. *Calendar:* Tokyo source sessions

Global risk and equities

Feature / final status	Definition and hypothesis
`vix` Retained	VIX_d Level of implied equity volatility as a stress-state hypothesis. *Calendar:* US source sessions
`vix_chg_1d` Retained	VIX_d - VIX_{d-1} Change in implied volatility; possible stress repricing. *Calendar:* US source sessions
`vix_chg_5d` Retained	VIX_d - VIX_{d-5} Change in implied volatility; possible stress repricing. *Calendar:* US source sessions
`spy_1d` Retained	ℓ_1(SPY) US equity performance as broad risk context. *Calendar:* US source sessions
`spy_5d` Retained	ℓ_5(SPY) US equity performance as broad risk context. *Calendar:* US source sessions

Feature / final status	Definition and hypothesis
`qqq_minus_spy_5d` Retained	ℓ_5(QQQ) - ℓ_5(SPY) Growth and technology leadership relative to the broad market. *Calendar:* US source sessions
`qqq_minus_spy_20d` Retained	ℓ_{20}(QQQ) - ℓ_{20}(SPY) Slower growth-versus-market rotation. *Calendar:* US source sessions
`smh_minus_qqq_5d` Retained	ℓ_5(SMH) - ℓ_5(QQQ) Semiconductor leadership relative to the technology-heavy index. *Calendar:* US source sessions

Dollar and relative yen performance

Feature / final status	Definition and hypothesis
`usd_basket_5d` Retained	$B_{5,t}$ = (-ℓ_5(EURUSD) - ℓ_5(GBPUSD) - ℓ_5(AUDUSD) + ℓ_5(USDCAD) + ℓ_5(USDCHF)) / 5 Custom equal-weight dollar proxy excluding yen; not the official broad dollar index. *Calendar:* Common USD/JPY FX endpoints
`yen_relative_5d` Retained	ret_5d - $B_{5,t}$ Relative performance under unit basket exposure; not a pure yen shock. Exactly collinear with its two components. *Calendar:* Common USD/JPY FX endpoints

Commodities

Feature / final status	Definition and hypothesis
`wti_5d_lagged` Retained	$\ell_5(WTI)_{d-1}$, released with observation d Oil-price context for an energy importer; spot observations, not a futures trading return. *Calendar:* EIA/FRED native observation sequence
`gld_5d` Retained	ℓ_5(GLD) Gold as a monetary and risk-context hypothesis. *Calendar:* US source sessions

Positioning and stress

Feature / final status	Definition and hypothesis
`cftc_net` Retained	q_j = (LF $longs_j$ - LF $shorts_j$) / open $interest_j$ Leveraged-fund yen positioning; positive means net long yen. *Calendar:* Distinct CFTC report sequence
`cftc_net_chg_4reports` Retained	q_j - q_{j-4} Building or unwinding positioning across reports; four reports need not mean four weeks. *Calendar:* Distinct CFTC report sequence

Feature / final status	Definition and hypothesis
`cftc_release_age_days` Retained	round(audit hours since release, 3) / 24 Age of published information, distinct from the age of the positions it describes. The existing formula uses audit hours rounded to three decimals before dividing by 24. *Calendar:* Decision rows / actual release timestamps
`pos_z` Retained	$z_{52\ \text{reports}}(q)_j$ Positioning relative to its own trailing report history. *Calendar:* Last 52 distinct CFTC reports
`vol_z` Retained	$z_{260}(\text{vol_20})_t$ Volatility relative to its own trailing decision history. *Calendar:* Last 260 decision rows
`short_yen_stress` Retained	$\max(0, -q_j) \times \max(0, \text{vol_z}_t)$ Observed net short-yen exposure during above-normal volatility; not total market carry leverage. *Calendar:* Aligned reports × decision rows

Calendar and events

Feature / final status	Definition and hypothesis
`event_in_window` Retained	**1**{a scheduled event falls in (17:00 t, 17:00 t+1]} Scheduled-news exposure; contains no announcement result. *Calendar:* Known event schedules / decision intervals

Additional interaction

Feature / final status	Definition and hypothesis
`gapchg_x_vixchg` Retained	$z_{260}(\text{gap2y_chg_5d})_t \times z_{260}(\text{vix_chg_5d})_t$ Conditional joint rate and volatility state as a policy/risk hypothesis. *Calendar:* Last 260 decision rows

7. Detailed saved results

Development A

Saved eight-quarter results. SD is sample dispersion across quarters, not a confidence interval.

Procedure	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Mean	SD	Pool
Logistic: own price	0.604	0.471	0.626	0.627	0.471	0.567	0.505	0.579	0.556	0.065	0.537
LightGBM: selected	0.543	0.525	0.606	0.542	0.520	0.552	0.489	0.491	0.534	0.037	0.518
XGBoost: selected	0.452	0.604	0.599	0.525	0.509	0.611	0.467	0.477	0.531	0.066	0.523
CatBoost: selected	0.647	0.586	0.550	0.488	0.584	0.513	0.523	0.489	0.547	0.055	0.507
LightGBM: own price (unmatched)	0.548	0.526	0.551	0.514	0.519	0.601	0.496	0.574	0.541	0.034	0.512
Constant prevalence	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.000	0.466

Development B

Saved eight-quarter results. SD is sample dispersion across quarters, not a confidence interval.

Procedure	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Mean	SD	Pool
Logistic: own price	0.604	0.471	0.626	0.627	0.471	0.567	0.505	0.579	0.556	0.065	0.537
LightGBM: selected	0.455	0.469	0.596	0.459	0.627	0.421	0.471	0.443	0.493	0.076	0.471
XGBoost: selected	0.554	0.536	0.520	0.495	0.582	0.630	0.430	0.487	0.529	0.061	0.516
CatBoost: selected	0.437	0.406	0.567	0.467	0.674	0.502	0.430	0.384	0.483	0.096	0.468
LightGBM: own price	0.586	0.441	0.498	0.516	0.580	0.634	0.430	0.557	0.530	0.072	0.525
XGBoost: own price	0.457	0.382	0.611	0.497	0.499	0.532	0.543	0.536	0.507	0.067	0.498
CatBoost: own price	0.438	0.462	0.476	0.507	0.616	0.495	0.371	0.582	0.493	0.078	0.480
Constant prevalence	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.000	0.466

Development C

Saved eight-quarter results. SD is sample dispersion across quarters, not a confidence interval.

Procedure	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Mean	SD	Pool
Logistic: own price	0.604	0.471	0.626	0.627	0.471	0.567	0.505	0.579	0.556	0.065	0.537
LightGBM: shortlist	0.497	0.585	0.459	0.512	0.559	0.587	0.538	0.502	0.530	0.045	0.502
XGBoost: shortlist	0.481	0.692	0.581	0.490	0.591	0.541	0.475	0.424	0.534	0.085	0.528
CatBoost: shortlist	0.584	0.605	0.552	0.492	0.535	0.505	0.582	0.500	0.544	0.044	0.531
LightGBM: own price	0.586	0.441	0.498	0.516	0.580	0.634	0.430	0.557	0.530	0.072	0.525
XGBoost: own price	0.457	0.382	0.611	0.497	0.499	0.532	0.543	0.536	0.507	0.067	0.498
CatBoost: own price	0.438	0.462	0.476	0.507	0.616	0.495	0.371	0.582	0.493	0.078	0.480
Constant prevalence	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.000	0.466

Original reserved sample and full-coverage sensitivity

Procedure	R1	R2	R3	R4	Mean	Pooled
XGBoost: shortlist	0.630	0.657	0.639	0.542	0.617	0.615
CatBoost: shortlist	0.606	0.645	0.677	0.460	0.597	0.593
LightGBM: shortlist	0.576	0.632	0.543	0.356	0.527	0.521
Logistic: own price	0.458	0.564	0.685	0.488	0.549	0.566
XGBoost: own price	0.583	0.610	0.583	0.434	0.553	0.552
CatBoost: own price	0.475	0.470	0.666	0.411	0.505	0.520
LightGBM: own price	0.502	0.629	0.594	0.442	0.542	0.532

Model	229: expanded	229: own	258: expanded	258: own
XGBoost	0.617	0.553	0.612	0.557
CatBoost	0.597	0.505	0.593	0.533
LightGBM	0.527	0.542	0.515	0.558

The primary 229-row comparison excludes 29 decisions because positioning observation age exceeds 35 days. The post-hoc 258-row sensitivity carries only reports actually published by the cutoff, preserves the original overlap and does not refit. R1 grows from 35/8 observations/positives to 64/19; later quarters are unchanged. Four event flags exist in the reserved year, but only one lies in the primary sample. These distinctions must remain attached to any reuse of the tables.

8. Full classification tables

The following saved pooled tables include both classes and macro/weighted averages. Development rows use each fold's frozen inner-chosen threshold; no threshold is fitted to pooled outer predictions. These are prediction summaries, not an execution backtest.

Run A development

517 outer-development predictions; 157 positives and 360 negatives. The pooled classifications apply each fold's own inner-chosen threshold, rather than fitting one threshold to the pooled outer predictions.

Model	TN	FP	FN	TP	BA	Accuracy
Logistic: own price	149	211	55	102	0.532	0.485
LightGBM: selected	183	177	74	83	0.518	0.515
XGBoost: selected	181	179	76	81	0.509	0.507
CatBoost: selected	181	179	76	81	0.509	0.507
LightGBM: own price (unmatched)	158	202	61	96	0.525	0.491
Constant prevalence	180	180	79	78	0.498	0.499

Model	Class/average	Precision	Recall	F1	Support
Logistic: own price	0	0.730	0.414	0.528	360
Logistic: own price	1	0.326	0.650	0.434	157
Logistic: own price	Macro	0.528	0.532	0.481	517
Logistic: own price	Weighted	0.608	0.485	0.500	517
LightGBM: selected	0	0.712	0.508	0.593	360
LightGBM: selected	1	0.319	0.529	0.398	157
LightGBM: selected	Macro	0.516	0.518	0.496	517
LightGBM: selected	Weighted	0.593	0.515	0.534	517
XGBoost: selected	0	0.704	0.503	0.587	360
XGBoost: selected	1	0.312	0.516	0.388	157
XGBoost: selected	Macro	0.508	0.509	0.488	517
XGBoost: selected	Weighted	0.585	0.507	0.527	517
CatBoost: selected	0	0.704	0.503	0.587	360
CatBoost: selected	1	0.312	0.516	0.388	157
CatBoost: selected	Macro	0.508	0.509	0.488	517
CatBoost: selected	Weighted	0.585	0.507	0.527	517
LightGBM: own price (unmatched)	0	0.721	0.439	0.546	360
LightGBM: own price (unmatched)	1	0.322	0.611	0.422	157
LightGBM: own price (unmatched)	Macro	0.522	0.525	0.484	517
LightGBM: own price (unmatched)	Weighted	0.600	0.491	0.508	517
Constant prevalence	0	0.695	0.500	0.582	360
Constant prevalence	1	0.302	0.497	0.376	157
Constant prevalence	Macro	0.499	0.498	0.479	517
Constant prevalence	Weighted	0.576	0.499	0.519	517

Run B development

517 outer-development predictions; 157 positives and 360 negatives. The pooled classifications apply each fold's own inner-chosen threshold, rather than fitting one threshold to the pooled outer predictions.

Model	TN	FP	FN	TP	BA	Accuracy
Logistic: own price	149	211	55	102	0.532	0.485
LightGBM: selected	195	165	90	67	0.484	0.507
XGBoost: selected	173	187	76	81	0.498	0.491
CatBoost: selected	106	254	52	105	0.482	0.408
LightGBM: own price	190	170	76	81	0.522	0.524
XGBoost: own price	247	113	114	43	0.480	0.561
CatBoost: own price	176	184	73	84	0.512	0.503
Constant prevalence	180	180	79	78	0.498	0.499

Model	Class/average	Precision	Recall	F1	Support
Logistic: own price	0	0.730	0.414	0.528	360
Logistic: own price	1	0.326	0.650	0.434	157
Logistic: own price	Macro	0.528	0.532	0.481	517
Logistic: own price	Weighted	0.608	0.485	0.500	517
LightGBM: selected	0	0.684	0.542	0.605	360
LightGBM: selected	1	0.289	0.427	0.344	157
LightGBM: selected	Macro	0.487	0.484	0.475	517
LightGBM: selected	Weighted	0.564	0.507	0.526	517
XGBoost: selected	0	0.695	0.481	0.568	360
XGBoost: selected	1	0.302	0.516	0.381	157
XGBoost: selected	Macro	0.499	0.498	0.475	517
XGBoost: selected	Weighted	0.576	0.491	0.511	517
CatBoost: selected	0	0.671	0.294	0.409	360
CatBoost: selected	1	0.292	0.669	0.407	157
CatBoost: selected	Macro	0.482	0.482	0.408	517
CatBoost: selected	Weighted	0.556	0.408	0.409	517
LightGBM: own price	0	0.714	0.528	0.607	360
LightGBM: own price	1	0.323	0.516	0.397	157
LightGBM: own price	Macro	0.518	0.522	0.502	517
LightGBM: own price	Weighted	0.595	0.524	0.543	517
XGBoost: own price	0	0.684	0.686	0.685	360
XGBoost: own price	1	0.276	0.274	0.275	157
XGBoost: own price	Macro	0.480	0.480	0.480	517
XGBoost: own price	Weighted	0.560	0.561	0.561	517
CatBoost: own price	0	0.707	0.489	0.578	360
CatBoost: own price	1	0.313	0.535	0.395	157
CatBoost: own price	Macro	0.510	0.512	0.487	517
CatBoost: own price	Weighted	0.587	0.503	0.523	517
Constant prevalence	0	0.695	0.500	0.582	360
Constant prevalence	1	0.302	0.497	0.376	157
Constant prevalence	Macro	0.499	0.498	0.479	517
Constant prevalence	Weighted	0.576	0.499	0.519	517

No-wrapper development

517 outer-development predictions; 157 positives and 360 negatives. The pooled classifications apply each fold's own inner-chosen threshold, rather than fitting one threshold to the pooled outer predictions.

Model	TN	FP	FN	TP	BA	Accuracy
Logistic: own price	149	211	55	102	0.532	0.485
LightGBM: shortlist	194	166	86	71	0.496	0.513
XGBoost: shortlist	203	157	78	79	0.534	0.545
CatBoost: shortlist	185	175	69	88	0.537	0.528
LightGBM: own price	190	170	76	81	0.522	0.524
XGBoost: own price	247	113	114	43	0.480	0.561
CatBoost: own price	176	184	73	84	0.512	0.503
Constant prevalence	180	180	79	78	0.498	0.499

Model	Class/average	Precision	Recall	F1	Support
Logistic: own price	0	0.730	0.414	0.528	360
Logistic: own price	1	0.326	0.650	0.434	157
Logistic: own price	Macro	0.528	0.532	0.481	517
Logistic: own price	Weighted	0.608	0.485	0.500	517
LightGBM: shortlist	0	0.693	0.539	0.606	360
LightGBM: shortlist	1	0.300	0.452	0.360	157
LightGBM: shortlist	Macro	0.496	0.496	0.483	517
LightGBM: shortlist	Weighted	0.573	0.513	0.532	517
XGBoost: shortlist	0	0.722	0.564	0.633	360
XGBoost: shortlist	1	0.335	0.503	0.402	157
XGBoost: shortlist	Macro	0.529	0.534	0.518	517
XGBoost: shortlist	Weighted	0.605	0.545	0.563	517
CatBoost: shortlist	0	0.728	0.514	0.603	360
CatBoost: shortlist	1	0.335	0.561	0.419	157
CatBoost: shortlist	Macro	0.531	0.537	0.511	517
CatBoost: shortlist	Weighted	0.609	0.528	0.547	517
LightGBM: own price	0	0.714	0.528	0.607	360
LightGBM: own price	1	0.323	0.516	0.397	157
LightGBM: own price	Macro	0.518	0.522	0.502	517
LightGBM: own price	Weighted	0.595	0.524	0.543	517
XGBoost: own price	0	0.684	0.686	0.685	360
XGBoost: own price	1	0.276	0.274	0.275	157
XGBoost: own price	Macro	0.480	0.480	0.480	517
XGBoost: own price	Weighted	0.560	0.561	0.561	517
CatBoost: own price	0	0.707	0.489	0.578	360
CatBoost: own price	1	0.313	0.535	0.395	157
CatBoost: own price	Macro	0.510	0.512	0.487	517
CatBoost: own price	Weighted	0.587	0.503	0.523	517
Constant prevalence	0	0.695	0.500	0.582	360
Constant prevalence	1	0.302	0.497	0.376	157
Constant prevalence	Macro	0.499	0.498	0.479	517
Constant prevalence	Weighted	0.576	0.499	0.519	517

Original eligible reserved sample

229 common eligible rows; 64 positives and 165 negatives. Each procedure uses its frozen inner-chosen threshold throughout the reserved year.

Model	TN	FP	FN	TP	BA	Accuracy
Logistic: own price	54	111	19	45	0.515	0.432
LightGBM: shortlist	88	77	30	34	0.532	0.533
XGBoost: shortlist	80	85	23	41	0.563	0.528
CatBoost: shortlist (nominee)	59	106	16	48	0.554	0.467
LightGBM: own price	52	113	14	50	0.548	0.445
XGBoost: own price	46	119	11	53	0.553	0.432
CatBoost: own price	1	164	0	64	0.503	0.284
Constant prevalence	0	165	0	64	0.500	0.279

Model	Class/average	Precision	Recall	F1	Support
Logistic: own price	0	0.740	0.327	0.454	165
Logistic: own price	1	0.288	0.703	0.409	64
Logistic: own price	Macro	0.514	0.515	0.431	229
Logistic: own price	Weighted	0.614	0.432	0.441	229
LightGBM: shortlist	0	0.746	0.533	0.622	165
LightGBM: shortlist	1	0.306	0.531	0.389	64
LightGBM: shortlist	Macro	0.526	0.532	0.505	229
LightGBM: shortlist	Weighted	0.623	0.533	0.557	229
XGBoost: shortlist	0	0.777	0.485	0.597	165
XGBoost: shortlist	1	0.325	0.641	0.432	64
XGBoost: shortlist	Macro	0.551	0.563	0.514	229
XGBoost: shortlist	Weighted	0.651	0.528	0.551	229
CatBoost: shortlist (nominee)	0	0.787	0.358	0.492	165
CatBoost: shortlist (nominee)	1	0.312	0.750	0.440	64
CatBoost: shortlist (nominee)	Macro	0.549	0.554	0.466	229
CatBoost: shortlist (nominee)	Weighted	0.654	0.467	0.477	229
LightGBM: own price	0	0.788	0.315	0.450	165
LightGBM: own price	1	0.307	0.781	0.441	64
LightGBM: own price	Macro	0.547	0.548	0.445	229
LightGBM: own price	Weighted	0.653	0.445	0.448	229
XGBoost: own price	0	0.807	0.279	0.414	165
XGBoost: own price	1	0.308	0.828	0.449	64
XGBoost: own price	Macro	0.558	0.553	0.432	229
XGBoost: own price	Weighted	0.668	0.432	0.424	229
CatBoost: own price	0	1.000	0.006	0.012	165
CatBoost: own price	1	0.281	1.000	0.438	64
CatBoost: own price	Macro	0.640	0.503	0.225	229
CatBoost: own price	Weighted	0.799	0.284	0.131	229
Constant prevalence	0	0.000	0.000	0.000	165
Constant prevalence	1	0.279	1.000	0.437	64
Constant prevalence	Macro	0.140	0.500	0.218	229
Constant prevalence	Weighted	0.078	0.279	0.122	229

9. Reproduction and implementation references

The source modules and selected command-line entry points are preserved from the audited refactor. The public exhibit builder reads aggregate CSV/JSON only. It does not acquire market data, reconstruct the feature matrix, load frozen models or rerun the 63 historical equivalence checks. Full reproduction needs the external research artifacts listed in the README and data-source inventory. A fresh download can differ from the historical snapshot; an unseeded historical solver can also differ on refit.

AI-assisted source inspection, derivation review, execution and drafting formed part of the workflow described in the main report. The references below document specific estimators and implementation conventions. Provider sources and access routes are listed separately in `data_sources.md`. Proprietary lecture/exam materials, signed forms, private market observations and credentials are omitted.

References

[1] CatBoost developers. CatBoost documentation: Common training parameters; score functions; transforming categorical features to numerical features. `https://catboost.ai/docs/en/references/training-parameters/common`, `https://catboost.ai/docs/en/concepts/algorithm-score-functions`, `https://catboost.ai/docs/en/concepts/algorithm-main-stages_cat-to-numberic`, 2026. Installed version 1.2.10; accessed 27 October 2026.

[2] CatBoost developers. CatBoost v1.2.10 cpu scoring and leaf-estimation source. `https://github.com/catboost/catboost/tree/v1.2.10/catboost`, 2026. score_calcers.h/.cpp, short_vector_ops.h, leafwise_scoring.cpp and online_predictor.h. Release-source check, 27 October 2026.

[3] LightGBM developers. LightGBM 4.7.0 documentation: Parameters and features. `https://lightgbm.readthedocs.io/en/v4.7.0/Parameters.html`, `https://lightgbm.readthedocs.io/en/v4.7.0/Features.html`, 2025. Version 4.7.0; accessed 27 October 2026.

[4] LightGBM developers. LightGBM v4.7.0 source: `src/treelearner/feature_histogram.hpp` and `serial_tree_learner.cpp`. `https://github.com/microsoft/LightGBM/tree/v4.7.0`, 2025. Release source checked 27 October 2026; leaf weights, gain and growth order.

[5] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems 31*, pages 6638–6648, 2018. arXiv:1706.09516; equations (4) and (5), Algorithms 1 and 2 and Section 5 used.

[6] Brian C. Ross. Mutual information between discrete and continuous data sets. *PLoS ONE*, 9(2):e87357, 2014.

[7] scikit-learn developers. Logistic regression (user guide, section 1.1.11) and LogisticRegression api reference, scikit-learn 1.9.1. `https://scikit-learn.org/stable/modules/linear_model.html`, 2026. Installed version checked 27 October 2026; mean-loss and penalty conventions.

[8] scikit-learn developers. `mutual_info_classif`, scikit-learn 1.9.1 (source `sklearn/feature_selection/_mutual_info.py`), 2026. Installed source inspected 27 October 2026.

[9] XGBoost developers. XGBoost documentation: XGBoost parameters and introduction to boosted trees. `https://xgboost.readthedocs.io/en/stable/parameter.html`, `https://xgboost.readthedocs.io/en/stable/tutorials/model.html`, 2026. Installed version 3.4.1; accessed 27 October 2026.

[10] XGBoost developers. XGBoost source: `src/tree/param.h`, `src/tree/driver.h`, `src/tree/split_evaluator.h`, `src/tree/hist/evaluate_splits.h`. `https://github.com/dmlc/xgboost/tree/v3.4.1`, 2026. Release v3.4.1 checked 27 October 2026: regularised leaf weights, unhalved gain, split thresholds and depth-wise expansion.